Median Filtered Image Restoration and Anti-Forensics Using Adversarial Networks

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***Abstract*—Median filtering is used as an anti-forensic technique to erase processing history of some image manipulations such as JPEG, resampling, etc. Thus, various detectors have been proposed to detect median filtered images. To counter these techniques, several anti-forensic methods have been devised as well. However, restoring the median filtered image is a typical ill-posed problem, and thus it is still difficult to reconstruct the image visually close to the original image. Also, it is further hard to make the restored image have the statistical characteristic of the raw image for the anti-forensic purpose. To solve this problem, we present a median filtering anti-forensic method based on deep convolutional neural networks, which can effectively remove traces from median filtered images. We adopt the framework of generative adversarial networks to generate images that follow the underlying statistics of unaltered images, significantly enhancing forensic undetectability. Through extensive experiments, we demonstrate that our method successfully deceives the existing median filtering forensic techniques.**

***Index Terms*—Anti-forensics, convolutional neural networks (CNNs), generative adversarial networks (GANs), median filtering (MF).**

I. Introduction

With the advent of social media such as Facebook, Instagram, Twitter, etc., and the universal use of digital cameras, people have uploaded many images on the Internet. On the other hand, as the use of photo-editing tools becomes easier, even nonexperts can forge images and damage the integrity of photos. Following this, many researchers have proposed the forensic techniques based on the assumption that the image tampering changes the inherent statistics of original images.

Median filtering (MF) is an operation widely used to remove outlying pixels such as salt-and-pepper noise and smooth regions while preserving edges. Due to this property, forgers may perform MF to hide traces of their manipulated images. As shown in [1], MF can be used in algorithms to prevent detection of other image operations in the manipulation process, such as resampling and JPEG compression, due to its nonlinearity. Therefore, it is of great importance to identify whether MF is applied to an image, as it implies that the image may have been modified by such image operations as well as MF itself. However, MF operation introduces undesired patterns called *streaking artifacts* (or blotches) [2], degrading image quality and leaving traces. Most of the existing MF detectors are designed to identify the presence of these artifacts.

Kirchner and Fridrich [3] employed *streaking artifacts* and *subtractive pixel adjacency matrix* (SPAM) [4] features to detect MF images. Cao *et al.* [5] used the probability of zero values on the first-order difference map in texture regions as a statistical fingerprint feature. Yuan [6] proposed a 44-D *median filtering forensics* (MFF) feature, computed from every pixel in overlapping blocks, to examine the distribution of the block median. Kang *et al.* [7] used ten coefficients of the autoregressive model for *median filtered residual* (MFR) as a feature. Chen *et al.* [8] proposed *global and local feature* (GLF) set composed of global probability features and the local correlation feature vector. Zhang *et al.* [9] presented the second-order *local ternary pattern* (LTP) to investigate MF from the angle of microtexture structures. Recently, forensic techniques using convolutional neural networks (CNNs) have been proposed, which do not require any hand-crafted feature engineering [10]–[12].

On the other hand, there have been many studies for countering MF forensic techniques. Fontani and Barni [13] pioneered anti-forensics of MF, which made MF images less detected while maintaining their peak signal-to-noise ratio (PSNR). Wu *et al.* [14] modified pixel value differences of MF images, which were modeled as a generalized Gaussian distribution. Dang-Nguyen *et al.* [15] proposed a technique that added noise to more textured regions of an MF image to evade being detected. Fan *et al.* [16] proposed an optimization framework using the pixel value difference as a prior for image enhancement and anti-forensics of the MF image. Sharma *et al.* [17] presented an anti-forensic attack designed as an optimization problem constrained by a difference of anisotropic and isotropic total variation regularization when given both an MF image and its raw one.

In this work, we propose an anti-forensic framework of MF using a generative adversarial network (GAN) [18] comprised of two networks: a generator and a discriminator. It is appropriate to use the GAN for the anti-forensic purposes in the

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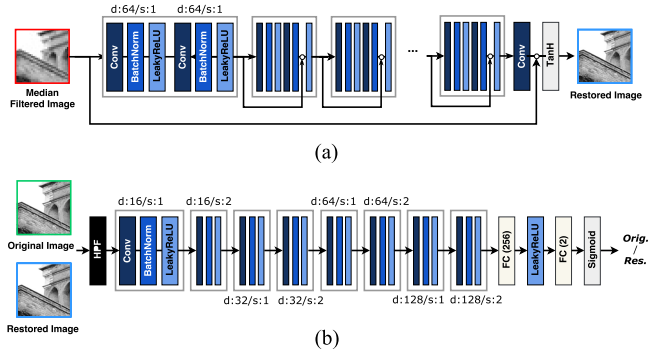


Fig. 1. Proposed network architectures for MF image restoration and anti-forensics. (a) Generator. (b) Discriminator. *d* and *s* denote the depth and stride of each convolutional layer, respectively.

following two aspects: 1) the configuration of the GAN resembles a forensic field, where forgers and forensic investigators are in opposition to each other; and 2) in general, the ultimate goal of a GAN is to obtain a strong generator whose output fools a discriminator. Also, the CNN-based generator is suitable for visual and statistical reconstruction of MF images with proper loss functions. Our contributions are summarized as follows.

1) Unlike [19], we propose a standalone machine-learning-based anti-forensics, which does not need any prior knowledge on forensic algorithms.
2) We train a network to generate anti-forensically modified (AFM) images similar to unaltered ones in statistical aspect, beyond visual restoration.
3) We perform extensive experiments against various kinds of MF detectors to demonstrate the effectiveness of our approach.

II. Anti-Forensic Networks for MF

We aim to restore an MF image so that it follows the statistical properties of a raw image beyond its simple visual restoration. To achieve this, we employ two networks in the form of a GAN [18]: a generator and a discriminator. We simultaneously train the two networks by optimizing the following minimax problem:

$$\min_{\theta_G}\max_{\theta_D}\mathbb{E}_{\mathbf{x}\sim p_{\text{raw}}(\mathbf{x})}\left[\log D_{\theta_D}(\mathbf{x})\right] + \mathbb{E}_{\mathbf{x}'\sim p_{\text{MF}}(\mathbf{x}')}\left[\log\left(1-D_{\theta_D}\left(G_{\theta_G}(\mathbf{x}')\right)\right)\right] \tag{1}$$

where θ_G and θ_D are learnable parameters for the networks G and D, respectively. The generator G transforms an MF image $\mathbf{x}'$ to make it follow the raw image distribution p_{raw}. Meanwhile, the discriminator D distinguishes the restored image samples from the raw ones.

A. Architecture of the Proposed Networks

1) Generator Network: We describe the details of the proposed networks. Inspired by [20], in our generator network, we use identical residual blocks as shown in Fig. 1(a). Each block consists of two 3×3 convolutional layers with 64 output filters, each of which is followed by batch normalization (BatchNorm) and leaky rectified linear unit (LeakyReLU) as an activation function. LeakyReLU shows better performance than ReLU in CNNs [21], and it is good to use it in the GAN framework for the stability in training [22]. Again, we place skip connections of block outputs, which are summed with the second BatchNorm output of each adjacent block. Also, we add the network input to the output of the last convolution layer, which is a sort of residual learning strategy resulting in fast and stable training [23].

2) Discriminator Network: A network with a common structure cannot even distinguish between simple MF images and unaltered ones due to interference with information irrelevant to trace of the processing [10]. Considering that the low-frequency part of MF images is similar to that of original ones, we make the discriminator focus on the high-frequency region to capture the statistical difference between restored images and raw ones. Specifically, we construct the first layer of the network by stacking the MFR and the filtered outputs with four high-pass kernels [24], the two first-order derivative kernels in horizontal and vertical directions, and the second-order derivative filter, which are computed from the network input and are zero-padded.

MFR used in the MF detections [7], [10] is defined as follows:

$$\mathbf{d} = \text{MF}_w(\mathbf{x}) - \mathbf{x} \tag{2}$$

where MF_w indicates $w\times w$ MF and can be regarded as a pooling layer, i.e., $w\times w$ median pooling with a stride of 1. It can be seen that max operation of max pooling used in the existing CNNs is replaced by median operation, which allows us to use it immediately in our discriminator network. Instead of applying the MF directly, however, we use the generator's inputs, or MF images, $\mathbf{x}'$ to avert sparse gradients.[1] Given a raw image $\mathbf{x}$ and the final generator G, we assume $\mathbf{x}' = \text{MF}_w(\mathbf{x}) \approx \text{MF}_w(G(\mathbf{x}'))$. To find underlying statistics of unaltered images beyond fooling MF detectors, we also employ other various kinds of high-pass filters. KV kernel represses image content and preserves high-frequency part, which is mainly used in steganalysis [24] and camera identification [25]. The other high-pass filters complement KV kernel.

In the remaining parts of the network, we use 3×3 convolutional layers with stride 1 and stride 2 in an alternating manner to reduce the dimensionality while avoiding sparse gradients [22]. After each convolution with stride 2, meanwhile, we increase the number of filters by a factor of 2. BatchNorm and LeakyReLU are placed after every convolutional layer.

B. Loss Functions

Our ultimate goal is to make the reconstructed image have statistical characteristics of the raw image with almost the same visual quality as before applying the median filter. From the view of training the generator network, we can model the loss function as follows:

$$\mathcal{L} = \lambda_{\ell_1}\mathcal{L}_{\ell_1} + \lambda_{\text{vgg}}\mathcal{L}_{\text{vgg}} + \lambda_{\text{adv}}\mathcal{L}_{\text{adv}} \tag{3}$$

where $\mathcal{L}_{\ell_1}$, $\mathcal{L}_{\text{vgg}}$, and $\mathcal{L}_{\text{adv}}$ represents pixelwise ℓ_1 loss, perceptual loss using the pre-trained VGG-16 network [26], and

[1] Note that our goal is not to find the pixel distribution of the MF image itself but to discover that of MFR.

adversarial loss, respectively. λ_X indicates a predefined weight term of loss $\mathcal{L}_X$, which determines the actual network configuration. In the following, we describe how the loss terms are defined and why they are needed.

For the image restoration task, in general, ℓ_1 loss can bring better visual quality than ℓ_2 loss [27]. Given an original image $\mathbf{x}$ of size $W \times H$ and its corresponding MF image $\mathbf{x}'$, the pixelwise ℓ_1 loss is calculated as follows:

$$\mathcal{L}_{\ell_1} = \frac{1}{N}\sum_{i=1}^{N} |\mathbf{x}_i - G(\mathbf{x}')_i| \tag{4}$$

where the subscript i refers to pixel location and $N = W \times H$. $G(\cdot)$ denotes the output of the generator. Some splotchy artifacts can be reduced in textured regions because ℓ_1 loss does not overpenalize larger errors. Nevertheless, if we depend only on the pixelwise loss, which measures low-level pixel information, the generator cannot properly represent textured regions, falling into local optima. The feature reconstruction loss for the lower layers, called perceptual loss, in the VGG-16 network helps produce an image that is visually more convincing, which is defined as ℓ_2 distance between feature representations of input image and target one [28]:

$$\mathcal{L}_{\text{vgg}} = \frac{1}{N}\sum_{i=1}^{N} \|\phi(\mathbf{x})_i - \phi(G(\mathbf{x}'))_i\|_2^2 \tag{5}$$

where $\phi(\cdot)$ represents the feature map, which is the activation of the second layer in the VGG-16 network. The perceptual loss results in quite plausible textures effectively. However, it has a limitation in the reconstruction of complex regions by trying to represent minute noise and leaving some artifact. For the generator to reduce less natural representation and make images with statistics of raw image as possible, we introduce adversarial loss, which makes the generator deceive the discriminator

$$\mathcal{L}_{\text{adv}} = \log\left(1 - D(G(\mathbf{x}'))\right) \tag{6}$$

where $D(\cdot)$ indicates the output of the discriminator.

III. Experimental Results

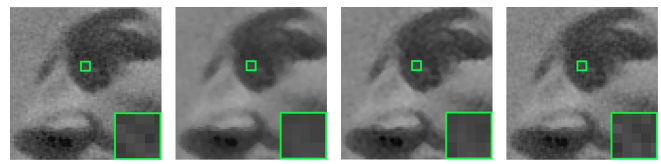
A. Experimental Setup

We implemented the proposed networks using the TensorFlow framework [29] and trained them on a machine equipped with a NVIDIA GTX 1070. In all experiments, we employed RAISE-1k [30], a large dataset of 1000 high-resolution (4928×3264) raw images, which had been created for digital image forensics, and another dataset composed of 200 raw images of size 3872×2592. We converted each original images to grayscale and cropped three adjacent nonoverlapped subimages with the size of 512×512 from the center of the high-resolution images.[2] In total, we created a set of 3600 images. Among them, we used 1500 images for training network and detector. To learn the networks, again, we randomly extracted 512000 image patches of size 40×40 from those images. Without loss of generality, we conducted anti-forensics (or restoration) of MF images obtained by the 3×3 median filter ($w = 3$). We compared with MF $\mathcal{M}$ (baseline) and the state-of-the-art anti-forensic method [16] $\mathcal{F}^f$, which did not require original images when generating forgeries. The set of hyperparameters $(\lambda, \gamma, \omega)$ was set to $(1500, 500, 0.1)$ which the authors had used for anti-forensic purpose.

TABLE I
Average PSNR and SSIM Values

Method	oPSNR (dB)	oSSIM	mPSNR (dB)	mSSIM
$\mathcal{M}$	40.5505	0.9550	–	–
$\mathcal{F}^f$ [16]	37.8684	0.9356	40.6966	0.9634
$\mathcal{F}$	**42.5061**	**0.9754**	**40.9868**	**0.9636**

oPSNR and oSSIM are calculated using the original image as the reference, while mPSNR and mSSIM are for those taking the MF image as the reference.



(a) (b) (c) (d)

Fig. 2. Reconstruction example ($4\times$ upsampled). (a) Original. (b) MF image. (c) Fan *et al.* [16]. (d) Proposed method. The bottom right on each image is a zoomed view of the area marked with a green box. Check the online version of this paper for the image detail.

B. Training Networks

Our generator network has eight identical residual blocks. Considering the limited GPU memory, we set the batch size to 64 images for training the generator. Before training the actual GAN to avoid unwanted local minima, we first trained the generator network with a learning rate of 5×10^{-4} and the weight terms $\lambda_{\ell_1} = 1.0$, $\lambda_{\text{vgg}} = 0.0$, and $\lambda_{\text{adv}} = 0.0$ during one epoch.

Next, the generator was trained with the weights $\lambda_{\ell_1} = 1.0$, $\lambda_{\text{vgg}} = 5 \times 10^{-6}$, and $\lambda_{\text{adv}} = 10^{-2}$ during 20 epochs.[3] On the other hand, we trained the discriminator with 128 images as a batch, consisting of 64 generated images and their corresponding raw ones. The leak slopes of LeakyReLUs were set to 0.2, which were used in the two networks. We used Adam [31] as optimizers with $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 10^{-8}$ for both the generator and discriminator networks. We exploited the fixed learning rate 5×10^{-4} and 5×10^{-6} for the generator and the discriminator, respectively.

C. Performance Evaluation

1) Image Restoration: We first conducted experiments to determine how well the proposed network restores the MF image. As in [16], we compared the restored images with the original and MF ones as references. Table I and Fig. 2 demonstrate that our approach $\mathcal{F}$ recovers the details removed by the MF process

[2] We followed the same experimental protocol and notations presented in [16]. For more details, refer to the paper.

[3] Considering the magnitudes of the converged losses, we set the parameters so that each loss has a balanced effect on the network training.

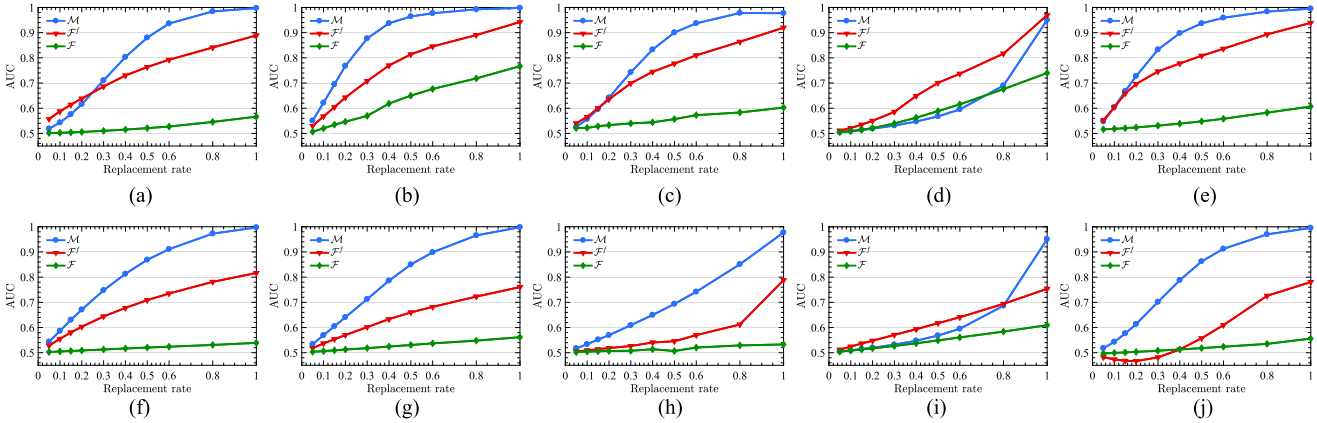


Fig. 3. AUC values according to different replacement rates against the SVM-based detectors trained under the *worst-case* (first row) and *optimistic* scenario (second row). (a) and (f) K_{SPAM} [4]. (b) and (g) K_{MFF} [6]. (c) and (h) K_{GLF} [8]. (d) and (i) K_{AR} [7]. (e) and (j) K_{LTP} [9].

TABLE II
ANTI-FORENSIC PERFORMANCES (AUC VALUES) AGAINST THE FOUR SCALAR-BASED DETECTORS

Method	K_K [3]	$\hat{K}_K$ [3]	K_C [5]	K_Y [6]
$\mathcal{M}$	0.7911	0.8950	0.9690	0.8888
$\mathcal{F}^f$ [16]	0.6543	0.6376	0.7728	0.6562
$\mathcal{F}$	**0.5030**	**0.5180**	**0.5958**	**0.5467**

TABLE III
ANTI-FORENSIC PERFORMANCES (AUC VALUES) AGAINST THE CNN-BASED MF DETECTORS

Method	Chen *et al.* [10]	Bayar and Stamm [11]
$\mathcal{M}$	0.9954	0.9746
$\mathcal{F}^f$ [16]	0.9499	0.8817
$\mathcal{F}$	**0.5398**	**0.5608**

well with higher average oPSNR, oSSIM, mPSNR, and mSSIM than $\mathcal{F}^f$. On the other hand, $\mathcal{F}^f$ even distorts the MF images and leads to lower average oPSNR and oSSIM than $\mathcal{M}$.

2) Anti-Forensics of MF: We considered various type of MF detectors for evaluation of anti-forensic performance, i.e., scalar-based detectors [3], [5], [6], support vector machine (SVM)-based detectors [4], [6]–[9], and CNN-based detectors [10], [11]. In all experiments, we used the same number of positive samples (MF or AFM images) and negative samples (unaltered images).

For scalar-based detectors, we verify their forensic performance as to how well they distinguish each forged image and its corresponding unaltered one. Table II shows the result. Note that the closer to 0.5 the area under the curve (AUC) value is, the more forensically undetectable the forged image is.

For SVM-based detectors, we follow Fan *et al.* [16], in which they considered two scenarios, i.e., *worst-case* and *optimistic* cases. For a given unaltered image, we replace the square central region of the image with its corresponding MF or AFM image by a replacement rate $r \in \{0.05, 0.1, 0.15, 0.2, 0.3, 0.4, 0.5, 0.6, 0.8, 1\}$. Under the worst-case scenario, an SVM-based detector is trained for each kind of features extracted from MF (or AFM) images at each replacement rate. As can be seen in Fig. 3(a)–(e), the proposed method has the lowest slope in almost all the cases. In other words, it means the highest undetectability. In the optimistic scenario, MF detectors are trained using original and MF images without any center replacement. In this case, therefore, the proposed method easily fools the detectors with AUC values around 0.5, as shown in Fig. 3(f)–(j). On the other hand, Fan *et al.* [16] show somewhat lower performance because the authors did not consider the smooth region, such as the scene of sky, in design.

CNN-based MF detectors are trained and used for classification on patches of a certain size, e.g., 64×64 [10] or 227×227 [11]. Therefore, we extracted nonoverlapped patches from the test image set according to the configuration of each network. Table III shows the AUC values calculated using the output of each network, i.e., softmax score. Our method made the detectors helpless very successfully in contrast to [16].

Although the proposed technique shows the best performance in most of the cases, it does not have perfect forensic undetectability, as shown in Fig. 3(b)–(d). This seems to be because the generator network is not good at restoring some noise, such as salt-and-pepper-like sparse noise, which does not leave traces after MF.

IV. CONCLUSION

In this paper, we presented an MF anti-forensic method based on two adversarial networks, which is composed of a generator and a discriminator. Unlike traditional anti-forensics methods, we achieved that goal in a data-driven way. We have shown through rigorous experiments that the proposed technique outperforms the state-of-the-art method regarding visual and statistical properties. In the future, we expect that our approach can be extended to anti-forensics of other image operations.

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